Quality Information Document

Ocean Monitoring Indicator (OMI) of the Copernicus Marine Environment Monitoring Service

Global yearly ocean CO_2 sink

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I INTRODUCTION

I.1 Product name

GLOBAL_OMI_HEALTH_carbon_co2_flux_integrated

I.2 Indicator evaluation method

The OMI global yearly ocean CO_2 sink represents the ocean uptake of CO_2 from the atmosphere computed over the whole ocean. It is computed by multiplying the 12-month running mean of the surface ocean downward CO_2 mass flux (fgco2) [molC m-2 yr-1], a variable from the MULTIOBS_GLO_BIO_CARBON_SURFACE_REP_015_008 product, by the area of each grid cell, and then taking the sum over all cells. The unit is finally converted to PgC per year (Figure 1). The yearly estimate of the ocean CO_2 sink corresponds to the mean of a 100-member ensemble (see the method described in Chau et al. (2022, 2023) in the QUID for GLOBAL_OMI_HEALTH_carbon_ph_area_averaged). In line with Friedlingstein et al. (2022), yearly uncertainties are estimated as their corresponding ensemble standard deviation (1σ).

I.3 Copernicus Marine Environment Monitoring Service products used as input for the computations

Product name (and Type)	Documentation
MULTIOBS_GLO_BIO_CARBON_SURFACE_REP_015_008	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-MOB-QUID-015-008.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-MOB-PUM-015-008.pdf

II SCIENTIFIC EVALUATION

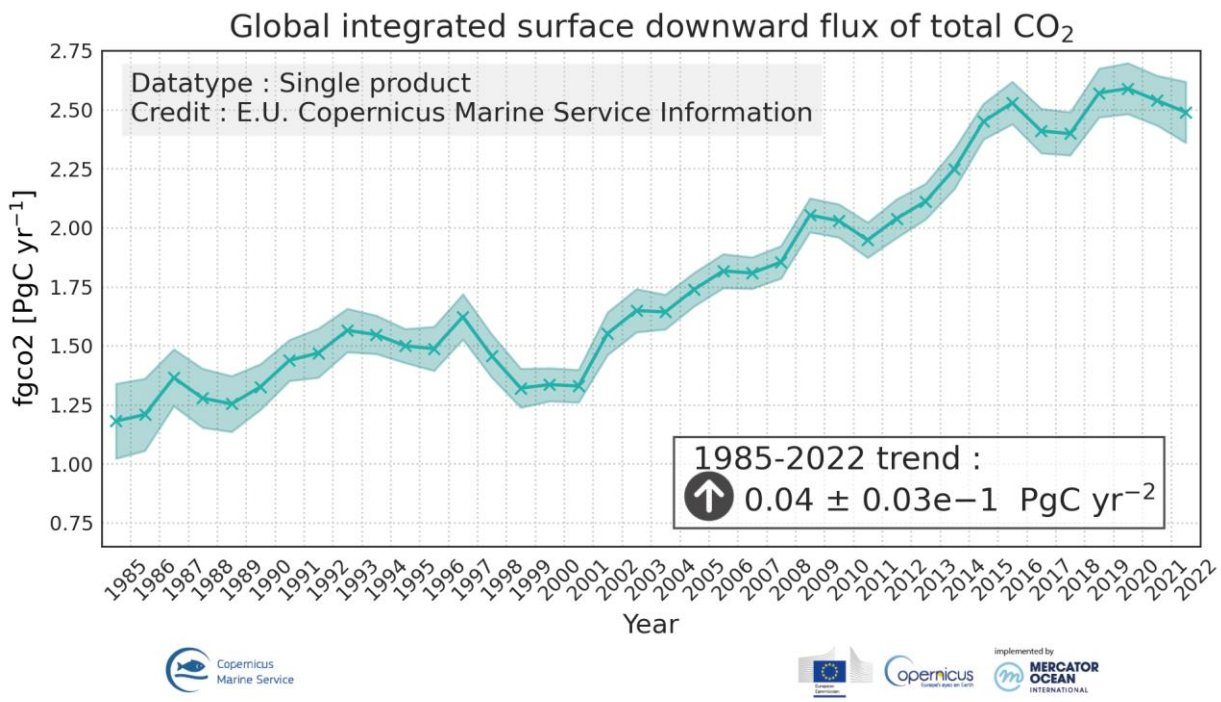

Figure 1: Area-integrated yearly surface downward flux of total CO_2 for 1985-2022. Plain line: mean of 100 samples; range: empirical 68 % confidence interval (mean estimate ± standard deviation) computed from 100 samples. Trend and uncertainty are defined as the slope and its residual standard deviation estimated with a linear least-squares regression.

The global ocean CO_2 sink corresponds to the global integral of fgco2 over space and time. The rate of uptake increases by 0.04±0.03e-1 PgC/yr^2 over the full period. The yearly flux time series presents a plateau in the 90s followed by an increasing growth rate of 0.06±0.04e-1 PgC/yr^2 since 2000. In 2022 [resp. 2021], the global ocean CO_2 sink was 2.49±0.13 [resp. 2.54±0.12] PgC/yr (Figure 1). The average of yearly contemporary fluxes over the full period 1985-2022 is 1.79±0.10 PgC/yr with an interannual variability (temporal standard deviation) of 0.45 PgC/yr.

The scientific evaluation relies on the comparison with estimates of the anthropogenic CO_2 uptake by the ocean that are published by the 2022 (GCB2022, Friedlingstein et al. 2022). The reconstructed field covers approximately 96.37% of the total ocean area (361.9e6 km^2). In order to compare the present estimates to Friedlingstein et al. (2022), contemporary fluxes are corrected for the total area and the preindustrial outgassing of riverine carbon by adding 0.65 PgC/yr (Regnier et al. 2022). This adjustment yields an anthropogenic flux estimate of 3.23±0.13 [resp. 3.28±0.11] PgC/yr for 2022 [resp. 2021]. This is at the upper end of the estimate by Friedlingstein et al. (2022) of 2.9±0.4 [resp. 2.9±0.4] PgC/yr.

The latter estimate corresponds to the mean of flux estimates from an ensemble of global ocean biogeochemical general circulation models (GOBMs) and an ensemble of data-based reconstruction methods. The GCB2022 uncertainty estimate combines random uncertainty (ensemble standard deviation of model- and data-based flux estimates) and systematic uncertainty (bias of model-based estimates in anthropogenic carbon cumulation). The new ensemble-based approach (Chau et al. 2022, 2023) yields spatio-temporal varying uncertainties which are lower than the previous published total estimate of ± 0.4 PgC/yr. The uncertainty estimates of the annual CO_2 uptake for the years in the 2000s are smaller than the uncertainties estimated for the years in the 1980s or recent years (Figure 1), as the sea surface spco2 observation network densifies. As shown in the following table, the somewhat higher anthropogenic flux estimates of the ensemble approach fall within the full range (±3 PgC/yr) of those by Friedlingstein et al. (2022). Further refinement of ocean CO_2 sink estimates relies on improved and sustained ocean observation efforts.

Periods	OMI_HEALTH_carbon_co2_flux_integrated [PgC/yr] (Mean ± uncertainty)		Friedlingstein et al. (2022) - anthropogenic flux estimate [PgC/yr]
	Contemporary flux estimate	Anthropogenic flux estimate (see text for definition)	
2022	3.13 ± 0.1	3.23±0.13	2.9±0.4
2021	2.54 ± 0.11	3.28±0.11	2.9±0.4
2012-2021	2.39 ± 0.09	3.23 ± 0.13	2.9±0.4
2000-2009	1.68 ± 0.08	2.39 ± 0.08	2.3±0.4
1990-1999	1.47 ± 0.09	2.18 ± 0.09	2.1±0.4
1985-1989	1.26 ± 0.14	1.95 ± 0.14	-

A full discussion regarding this OMI can be found in section 2.10 of the Ocean State Report 4 (Gehlen et al., 2020) and in Chau et al. (2022).

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